Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the classification of various classes using k-Nearest Neighbor approach	
Experiment No: 03	Date:	Enrollment No: 92301733043

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab

Theory:

This algorithm is used to solve classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and match new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

K-NN algorithm assumes the similarity between the new case/data and available cases and puts the new case into the category that is most like the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much like the new data.

Example: Suppose we have an image of a creature that looks like cat and dog, but we want to know whether it is a cat or dog. So, for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find similar features to the cats and dog's images and based on the most similar features it will put it in either cat or dog category.

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyze the dataset

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4. Pre-process the data

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5. Visualize the Data
6. Separate the feature and prediction value columns
7. Select the number K of the neighbors
8. Calculate the Euclidean distance of K number of neighbors
9. Take the K nearest neighbors as per the calculated Euclidean distance.
10. Among these k neighbors, count the number of data points in each category.
11. Assign the new data points to that category for which the number of the neighbor is maximum.
12. Our model is ready.

Pre Lab Exercise:

a. What is K in KNN-approach?

Ans: K represents the number of nearest data points that the algorithm considers making a prediction

b. Which are the types of distance metrices?

Ans: Euclidean Distance, Manhattan Distance, Minkowski Distance, etc.

c. What can be the criteria of selection of the value of K?

Ans: Avoid underfitting and overfitting, use cross validation, odd value of K

d. What are the advantages of KNN algorithm?

Ans: No training time, high adaptability to new data, non-linear, multi-class data without assuming underlying distributions

Program (Code):

```
import time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
```

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```
from sklearn.pipeline import Pipeline

from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix
)

digits = load_digits()
X = digits.data
y = digits.target
images = digits.images

plt.figure(figsize=(8, 4))
for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(images[i], cmap="gray")
    plt.title(f"{y[i]}")
    plt.axis("off")
plt.suptitle("Sample Digits")
plt.tight_layout()
plt.show()


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
knn_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("knn", KNeighborsClassifier())
])

knn_param_grid = {
    "knn__n_neighbors": [3, 5, 7, 9, 13, 15, 20]}
```

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```
rf_pipeline = Pipeline([
   ("rf", RandomForestClassifier(random_state=42))
])

rf_param_grid = {
   "rf__n_estimators": [25, 50, 100, 200, 500],
   "rf__max_depth": [None, 10, 20]}

mlp_pipeline = Pipeline([
   ("scaler", StandardScaler()),
   ("mlp", MLPClassifier(
      max_iter=1000,
      random_state=42,
      early_stopping=True
   ))
])

mlp_param_grid = {
   "mlp__hidden_layer_sizes": [(64,), (128,), (64, 64)],
   "mlp__activation": ["relu", "tanh"],
   #"mlp__alpha": [0.0001, 0.001],
   "mlp__learning_rate_init": [0.001, 0.01, 1.0, 5.0, 10.0]
}

#Training the model
def train_with_timing(pipeline, param_grid):
   start = time.time()
   grid = GridSearchCV(
      pipeline,
      param_grid,
      cv=5,
      scoring="f1_weighted",
      n_jobs=-1
   )
   grid.fit(X_train, y_train)
   train_time = time.time() - start
   return grid, train_time
```

```
knn_grid, knn_train_time = train_with_timing(knn_pipeline, knn_param_grid)
rf_grid, rf_train_time = train_with_timing(rf_pipeline, rf_param_grid)
mlp_grid, mlp_train_time = train_with_timing(mlp_pipeline, mlp_param_grid)

#Inferencing the model
def inference_time(model):
   start = time.time()
   model.predict(X_test)
   return time.time() - start

knn_test_time = inference_time(knn_grid.best_estimator_)
rf_test_time = inference_time(rf_grid.best_estimator_)
mlp_test_time = inference_time(mlp_grid.best_estimator_)

#Evaluation
def evaluate_model(model):
   y_pred = model.predict(X_test)
   return {
      "Accuracy": accuracy_score(y_test, y_pred),
      "Precision": precision_score(y_test, y_pred, average="weighted"),
      "Recall": recall_score(y_test, y_pred, average="weighted"),
      "F1-score": f1_score(y_test, y_pred, average="weighted")
   }

knn_metrics = evaluate_model(knn_grid.best_estimator_)
rf_metrics = evaluate_model(rf_grid.best_estimator_)
mlp_metrics = evaluate_model(mlp_grid.best_estimator_)

#Result with comparison
results_df = pd.DataFrame(
   [knn_metrics, rf_metrics, mlp_metrics],
   index=["KNN", "Random Forest", "MLP"]
)

results_df["Training Time (s)"] = [
   knn_train_time, rf_train_time, mlp_train_time
]

results_df["Inference Time (s)"] = [
```

```
    knn_test_time, rf_test_time, mlp_test_time
]

print(results_df)

results_df.iloc[:, :4].plot(kind="bar", figsize=(10, 6))
plt.title("Model Performance Comparison (Digits)")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.grid(axis="y")
plt.show()

models = {
    "KNN": knn_grid.best_estimator_,
    "Random Forest": rf_grid.best_estimator_,
    "MLP": mlp_grid.best_estimator_
}

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, (name, model) in zip(axes, models.items()):
    cm = confusion_matrix(y_test, model.predict(X_test))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", ax=ax)
    ax.set_title(name)
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")

plt.tight_layout()
plt.show()
```

Results:

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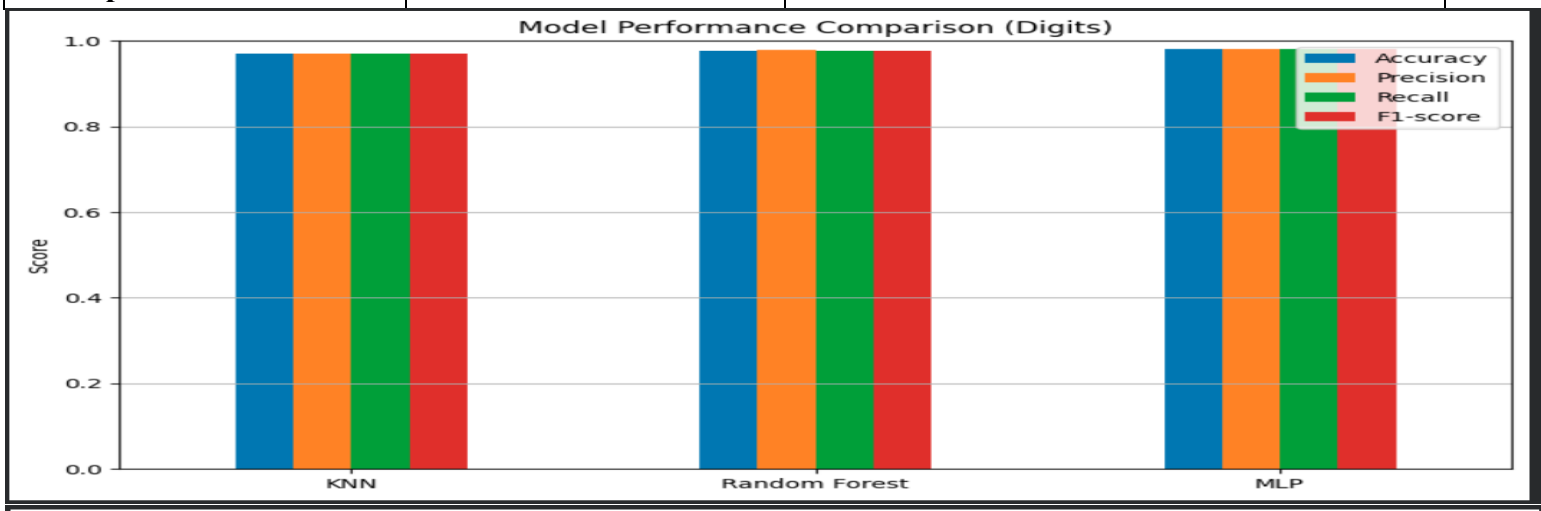

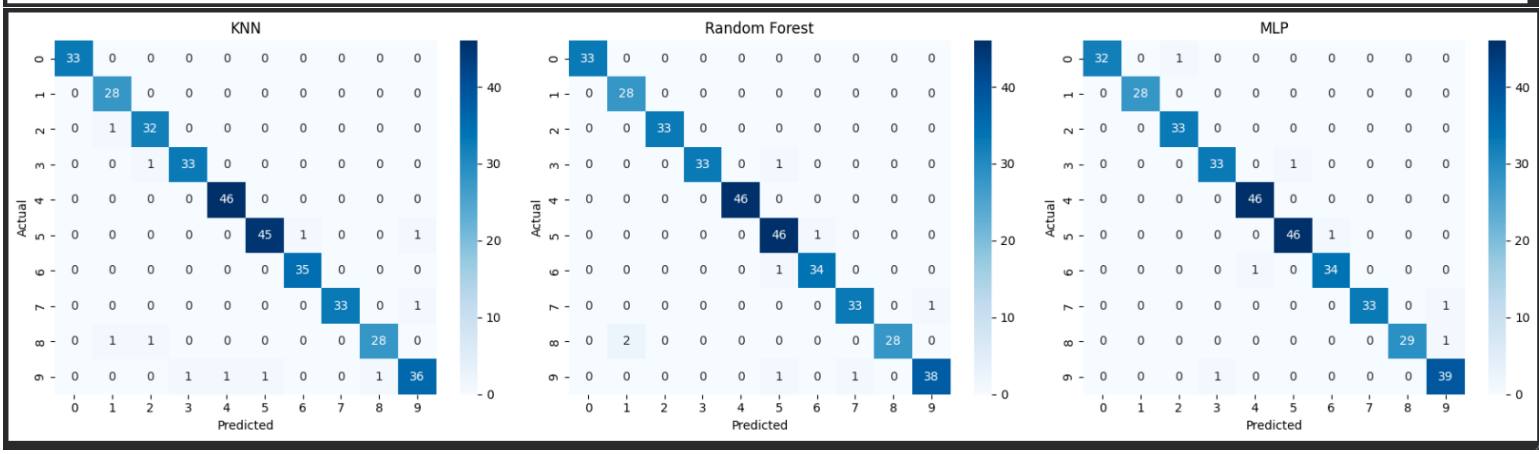

Observation and Result Analysis:

a. Nature of the dataset

Ans: The dataset is a supervised, multi-class classification dataset consisting of numerical pixel features representing handwritten digits (0–9).

b. During Training Process

Ans: During training, the dataset is used as a **supervised labeled dataset** where models learn patterns between pixel features and digit labels using training data and validate performance using cross-validation.

c. After the training Process

Ans: After training, the dataset is used as test data to evaluate the trained model's performance by making predictions and calculating metrics like accuracy, precision, recall, and F1-score.

d. Observation over the Learning Curve

Ans: Learning curve shows how model performance improves with training. It helps identify overfitting, underfitting, and the point where the model achieves stable and optimal performance.

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Post Lab Exercise:

a. Is KNN useful in regression-based problems?

Ans: Yes, KNN is useful in regression problems. It predicts output by taking the average of values of K nearest neighbors instead of majority voting.

b. Why KNN is a non-parametric algorithm?

Ans: KNN is a non-parametric algorithm because it does not assume any fixed data distribution or mathematical model and makes predictions directly using stored training data.

c. What are the assumptions of KNN approach?

Ans: KNN assumes that similar data points are located close to each other, distance measures correctly represent similarity, data is properly scaled, and sufficient training data is available for accurate prediction.

d. How can the KNN approach make the prediction of the unseen dataset?

Ans: KNN predicts unseen data by calculating distance from all training points, selecting the K nearest neighbors, and using majority voting (classification) or averaging (regression) to produce the final prediction.

e. Why is KNN called a Lazy Learner?

Ans: KNN is called a lazy learner because it does not build a model during training and performs all computations only when making predictions on new data.

f. Why should we not use KNN for large dataset?

Ans: KNN is not suitable for large datasets because it requires high computation time, large memory, and slow prediction since it must calculate distance from all training data points for each new input.